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Forecasting test cricket match outcomes in play

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ABSTRACT

This paper forecasts match outcomes in test cricket in play, session by session. Match outcome probabilities at the start of each session are forecast using a sequence of multinomial logistic regression models. These probabilities can assist a team captain or management in considering a certain aggressive or defensive batting strategy for the coming session. We investigate how the outcome probabilities (of a win, draw, or loss) and covariate effects vary session by session. The covariates fall into two categories, pre-match effects (strengths of teams, a ground effect, home field advantage, outcome of the toss) and in-play effects (score or lead, overs-used, overs-remaining, run-rate, and wicket resources used). The results indicate that the lead has a small effect on the match outcome early on but is dominant later; pre-match team strengths, ground effect and home field advantage are important predictors of a win early on; and wicket resources used remains important throughout a match.

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1. Introduction

Test cricket began in the late 1800s with matches which were not time-limited. In early matches, teams had an unlimited amount of time in which to chase a target or to bowl out their opponents twice. The batting strategy was very simple in timeless matches; a team would seek to score as many runs as possible and to utilize all of their batting resources. Only nine innings were ever declared in the ninety-nine timeless matches played. The fixed-duration format was universally adopted in 1939, and at the present day, test matches are played over five consecutive days with three sessions in a day. To win a test match, broadly speaking, a team needs to bowl their opponents out twice within the time limit of five days. The five-day time limit necessitates strategic play, particularly with respect to batting. A batting team needs to play each session according to a particular batting strategy in order to optimize the match outcome from their point of view. The optimal batting strategy in a given session will depend upon the current match situation or match state. The same is true for bowling. We use a range of covariates to summarise the match state quantitatively at each stage, then use the match state to forecast the match outcome. We anticipate that these forecasts could also be used by team captains to guide strategic choices. They might also be used for rating player contributions.

Some work has been completed on predicting match outcomes in test cricket: Brooks, Faff, and Sokulsky (2002) used an ordered probit model with batting and bowling strengths, and claimed to predict 71% of outcomes correctly. Scarf and Shi (2005) used logistic regression techniques to develop a model of match outcome probabilities, given the position at the end of the third innings. Scarf and Akhtar (2011) extended this work to the positions at the end of the first and second innings; their models have been used to consider the declaration strategy and the follow-on decision. Scarf, Shi, and Akhtar (2011) developed a model of the match outcome given the match state at some point during the third innings, and used this model to consider the batting strategy during the third innings. Our approach in this paper is new, in that we model the match outcome

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given the position at the start of each session. Match outcome probabilities are modelled using multinomial regression, with a win, draw, or loss response, and explanatory variables or covariates relating to the match state at the start of each session. These covariates include the lead, wicket resources of teams used, run-rate, a home advantage factor, and surrogates for the state of the pitch (ground effect) and the pre-match strengths of teams. We also investigate how the covariate effects vary from session to session. We attempt to compare our results (graphically) with bookmakers' odds by means of examples. This is illustrated in the form of a probability chart. Such charts may help management, players and fans to gain a better understanding of test cricket. Our study might also be generalized to other sports. The fitting of a sequence of multinomial regression models also raises methodological questions about how best to fit such models, which are beyond the scope of the present paper.

There is also other published work on test cricket that is related to our study here. Most recently, Lenten (2008) analysed the decline in the frequency of draws in test cricket over the last fifteen years. In a similar approach to ours, Allsopp and Clarke (2004) employed multinomial logistic regression to determine which factors affect final match outcomes in test cricket (win, draw or loss), at the stage when both teams had completed their first innings. Crowe and Middeldrop (1996) studied the rates of leg before wicket dismissals between Australian and visiting teams for test cricket series played in Australia between 1977 and 1994. Ringrose (2006) investigated leg before wicket dismissals in test cricket using generalized linear and mixed models. Clarke (1988) and Preston and Thomas (2000) investigated optimal batting strategies in one-day matches. However, test matches are different, because, in one-day matches, there is no notion of playing out the time remaining for a draw. In fact, de Silva and Swartz (1997) found that winning the toss does not provide a competitive advantage in a one-day match.

The remainder of the paper proceeds as follows. In the next section, the dataset of test match outcomes is briefly described, followed by a description of the modelling approach. The results of the model fitting are then presented, focussing on questions regarding which explanatory variables are important at each stage of the match, how strong their influence is, their forecasting abilities, and way in which a knowledge of the match outcome given the match position or state might be used to consider batting and bowling strategies. We conclude with a discussion of the limitations of our work, and opportunities for further development.

2. Data description

Our data were obtained from the EPSNcricinfo website (http://www.stats.cricinfo.com/ci/content/records/307847.html/; accessed on 10.10.10). Many forms of information are available on this website, including, for example, a ball-by-ball commentary, partnership scores, ground statistics, and session by session match summaries. An extract of the data used in our analysis is presented in Table 1. The complete dataset (146 matches) relates to all of the test matches in the period between November 2005 and March 2010, excluding those matches where the session by session data were not available or where more than 90 overs were lost to poor weather. Session by session information is not generally available prior to November 2005.

To account for team strengths, we use the International Cricket Council's official ratings (see http://icc-cricket.yahoo.net/match_zone/team_ranking.php; accessed on 28.10.10) and calculate the ratings differences (RD), the differences between the ICC ratings of the competing teams. We also use win percentage differences (W%D), the difference between the win percentages of the competing teams, as a covariate, and compare this to the ratings differences.

The number of wickets down is transformed into the wicket resources used. We define the wicket resources for wicket i ($i = 1, \ldots, 10$) to the total runs. Partnership data for 197 matches, collected by Scarf, Shi, and Akhtar (2011), were used for calculating the wicket resources (Table 2). We use wicket resources used (WR) as a covariate in our analysis. The Duckworth/Lewis method (Duckworth & Lewis, 1998, 2004), which is employed in rain-interrupted limited-overs matches, is based on the notion of resources, although in their analysis, resources are based on both overs remaining and wickets in hand, rather than only wickets in hand. Our notion of wicket resources is essentially the relative value of a partnership if it is completed, in terms of runs scored. The lead is calculated session by session throughout the match, with respect to the reference team, the team batting first. The ground effect is the proportion of matches on each ground which were drawn over the period from 1877 to 2010. This proportion is shown in Fig. 1 for some pitches. The ground effect may vary over time, but data to enable us to assess this were not available.

3. Modelling the match outcome

Scarf and Akhtar (2011), Scarf and Shi (2005) and Scarf, Shi, and Akhtar (2011) described a multinomial regression model for explaining match outcome probabilities, given an end of the innings position. We use the same model to consider the multinomial response (win, draw, loss) as a function of the match position at the start of each session. With the match outcome Y taking values $(1, 0, -1)$, to denote a win, draw, and loss respectively, covariates denoted by X, and taking a draw (0) as the reference category, this model assumes that Y has a multinomial distribution, that is $Y \sim MN(p_1, p_0, p_{-1}; \sum p_i = 1)$, where p_1, p_0 and p_{-1} represent the probability of a win, a draw and a loss, respectively, with

$$\left.\begin{aligned} p_1 &= \exp(\alpha_1 + \beta_1^T X)/\{1 + \exp(\alpha_1 + \beta_1^T X) \\ &\quad + \exp(\alpha_{-1} + \beta_{-1}^T X)\}, \\ p_0 &= 1/\{1 + \exp(\alpha_1 + \beta_1^T X) + \exp(\alpha_{-1} + \beta_{-1}^T X)\}, \\ p_{-1} &= \exp(\alpha_{-1} + \beta_{-1}^T X)/\{1 + \exp(\alpha_1 + \beta_1^T X) \\ &\quad + \exp(\alpha_{-1} + \beta_{-1}^T X)\}. \end{aligned}\right\}$$

This model is the nominal multinomial logistic regression model.

Table 1
Test match data (extract of 146 test matches, November 2005–March 2010). The variables included are as indicated, with the win percentage difference, rating difference, and match result (1 win, 0 draw, −1 loss) recorded from the point of view of the reference team (batting first). The number of overs remaining was estimated on the basis of an expected 90 overs per day (see http://www.lords.org/laws-and-spirit/laws-of-cricket/laws/). The only variable not shown is the number of overs bowled per session.

										At lunch—day 1							At tea—day 1						
Reference team	Opponents	Venue	Date	Percentage of drawn matches at venue	Win percentage difference	Home factor for reference team (yes = 1, no = 0)	If reference team win the toss then 1, else 0	Rating difference	Result	Reference team batting then 1, else 2	Runs in session	Wicket resources used (reference team)	Wicket resources used (opponents)	Run-rate	Overs-remaining in the match	Lead	Reference team batting then 1, else 2	Runs	Wicket resources used (reference team)	Wicket resources used (opponents)	Run-rate	Overs-remaining in the match	Lead
W	A	Hobart	17/11/2005	0.22	−50	0	1	−53	−1	1	49	24.0	0	1.88	424	49	1	124	62.6	0	2.25	395	124
W	A	Adelaide	25/11/2005	0.27	−50	0	1	−53	−1	1	71	37.5	0	2.63	423	71	1	194	51.1	0	3.46	394	194
I	SL	Delhi	10/12/2005	0.47	−2	1	1	13	1	1	68	24.0	0	3.78	432	68	1	164	37.5	0	3.15	398	164
A	SA	Perth	16/12/2005	0.19	25	1	1	28	0	1	96	11.1	0	3.69	424	96	1	175	37.5	0	3.30	397	175
I	SL	Ahmadabad	18/12/2005	0.50	−2	1	1	13	1	1	51	11.1	0	4.25	438	51	1	125	62.6	0	2.78	405	125
A	SA	Melbourne	26/12/2005	0.15	25	1	1	28	1	1	60	11.1	0	3.00	430	60	1	162	24	0	2.84	393	162
P	I	Faisalabad	21/01/2006	0.54	8	1	1	−12	0	1	137	24.0	0	3.81	414	137	1	248	51.1	0	3.88	399	248

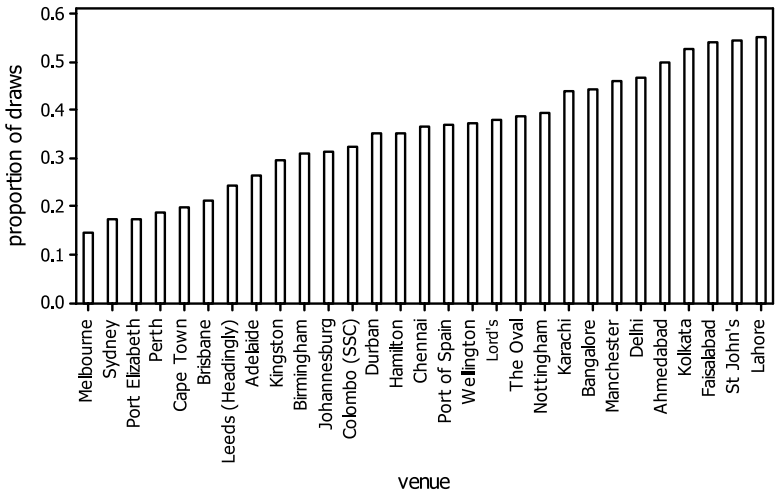

Fig. 1. Proportion of matches drawn by venue, for a selection of venues (1877–2010).

Table 2
Wicket resources and cumulative wicket resources used, as a function of the wicket partnership number, calculated from the partnership scores in 197 test matches over the period from February 1998 to June 2004 (all innings); expressed as percentages.

Wicket	1	2	3	4	5	6	7	8	9	10
Resources	12.24	12.13	13.53	14.55	11.48	10.99	8.16	7.17	5.24	4.51
Cumulative wicket resources used	12.24	24.37	37.90	52.45	63.93	74.92	83.08	90.25	95.49	100.00

The match outcome probabilities depend on the covariate vector X in different ways—through the coefficients β_1 and β_{-1} respectively. In a sense, the model is equivalent to fitting two binary logistic regression models, the first for the win-draw probability comparison and the second for the lose-draw probability comparison. In test cricket, match outcome categories can be ranked in the sense of win, draw and loss. However, ordinal logistic regression

requires an important assumption to be made, namely that the effect of the explanatory variables is the same for the different logit functions. This implies that each logit function has its own intercept term (α), but the same parameter (β) for each of the independent variables (e.g. β_1 for independent variable X_1). However, in test cricket, we have found that the win-draw comparison and the loss-draw comparison depend on the covariates in different ways—a property which is thus captured by the nominal but not the ordinal multinomial logistic regression model—and in our results (e.g. Table 6) one can see that the ordinal model provides a much poorer fit to the data. Also, if the natural ordering of the categories is ambiguous (e.g., a draw may be as good as a win for a team in the final match of a series when the team is leading the series 1–0), then it is more appropriate to consider a model for nominal outcomes (Dobson, 2002; Long, 1997). Other nominal multinomial regression models, such as probit regression, could be considered; however, we would expect the results to be very similar. This is because the logistic and probit functions are very nearly linearly related over the interval $0.1 \leq p \leq 0.9$ (McCullagh & Nelder, 1989, p. 109), and thus it is difficult to distinguish between the two functions based on goodness-of-fit. Ordinal (as opposed to nominal) probit regression has been used in football match prediction (e.g. Dobson & Goddard, 2003, and Koning, 2000).

In all, a total of fifteen such models, one for each session, are fitted.

To assess the model fit, we use the Akaike information criterion (AIC) (Sakamoto, Ishiguro, & Kitigawa, 1986) and Nagelkerke's R^2. This latter statistic is a modified form of the Cox and Snell (1989) pseudo R^2, and gives information about the explanatory power of the covariates in the model (Nagelkerke, 1991). Cox and Snell's pseudo R^2 is given by

$$R^2_{CS} = 1 - \exp\{-2(L_1 - L_0)/n\},$$

where L_1 represents the log-likelihood for the model with covariates, L_0 represents the log-likelihood for the model with no covariates, and n is the number of observations. Nagelkerke's R^2 is given by

$$R^2 = \frac{1 - \exp\{-2(L_1 - L_0)/n\}}{\{1 - \exp(2L_0)/n\}}.$$

The maximum value of Cox and Snell's R^2 is $1-\exp(2L_0)/n$, and therefore Nagelkerke's (modified) R^2 can vary from 0 to 1. Broadly speaking, Nagelkerke's R^2 for a generalized linear model is the proportion of variability in the outcome which is explained by the covariates in the model.

Rather than fit the sequence of regression models independently, one could consider sequential logistic regression in the manner of Elisheva, Noya, Yana, Dalit, and Benjamin (2000). Here, covariates in model i in the sequence which applies at time t_i are the linear predictor or score from model $i - 1$ which applies at time t_{i-1} plus covariates which relate to new information which has arisen between times t_{i-1} and t_i. This sequential approach is particularly useful when there are a large number of candidate covariates. Using the simpler approach, model selection can become a problem for models later in the sequence. Furthermore, using the score from a previous model as a covariate in a subsequent model implies that covariate effects in early models are still extant in later models, but with a down-weighted effect. On the other hand, the direct explanatory power of particular covariates can be calculated over the sequence of models using the simpler approach. The fitting of sequential models is not straightforward because an errors-in-variables approach is required, since the linear predictor is a random variable. We therefore leave the question of the relative merits of the two approaches as an open question.

4. Model fitting results

4.1. Start of the match

Initially, we model the match outcome at the start of a match. Outline model statistics for various fitted models are shown in Table 3. We consider a number of possible factors which may affect the outcome at the start of a match. To account for team strengths, we have used win percentage differences and the ICC rating differences. The rating difference (RD), ground effect (G) and home field advantage (H) were found to be important (based on the AIC and Nagelkerke's R^2). The rating difference has a better explanatory power than the win percentage difference. This may be because the ICC rating takes the result (win, draw, loss) into account, along with the win margin, wickets and opponent rating. The rating difference also has greater explanatory power than the ratio of the ICC ratings (R-ratio). Winning the toss was also considered in the model fitting, but was found to be unimportant. The ground effect (percentage of drawn matches) can be thought of as a surrogate for the general quality of batting conditions. The playing conditions vary from ground to ground and from country to country. For example, the playing conditions in Lahore at the Gaddafi Stadium are quite different to those in Leeds at Headingley. This is the reason why 55% of matches played at Gaddafi Stadium result in a draw, while the figure for Headingley is 24%. We find that this factor plays a very important role in the session by session analysis. We also note that Scarf and Akhtar (2011) and Scarf, Shi, and Akhtar (2011) failed to consider this as a covariate in their end of innings position models. A re-analysis of their declaration models indicates that the ground effect also has significant implications for match outcome prediction given the positions at the end of the first and second innings. However, the prediction given the end of third innings position is not affected, because by the end of the third innings the ground effect has broadly translated into the target and overs remaining.

Using the highlighted model in Table 3 (bold text), we can calculate the probability of a win, draw or loss given the position at the start of a match. This may help team captains and management to consider their batting and bowling strategies for the first session. Estimates for the highlighted model are shown in Table 4. Win and draw probabilities are presented in Fig. 2 for England when playing hypothetical matches against the rest of the current ICC member teams. It appears that the rating difference has a greater effect on the win probability than on the draw probability. The figure also shows that the

Table 3
Start of first session position: results of fitting the multinomial logistic regression model to 146 test match outcomes for various sets of predictors: log-likelihood, AIC and Nagelkerke's R^2. The covariates here are W%D, win percentage difference; RD, the ICC rating difference; H, home factor; G, ground effect; T, winning the toss; and R-ratio, ratio of the ICC ratings.

Model	Parameters	LL	AIC	Nag.R^2 (%)
W%D	4	−135.688	279.376	27.09
RD	4	−135.063	278.126	27.83
R-ratio	4	−135.903	279.806	26.84
RD + H	6	−130.362	272.724	33.17
RD + H + G	**8**	**−126.766**	**269.532**	**37.03**
RD + H + **G** + T	10	−126.549	273.098	37.25

Table 4
Start of first session position: fitted parameter estimates for a start of the match minimum AIC logistic regression model (nominal), with the covariates rating differences, home factor, and ground effect, with standard errors and *p*-values, 146 test matches (November 2005–March 2010).

		Coefficient	s.e.	*p*-value
Win/draw (1/0)	Intercept	1.369	0.670	0.041
	Rating difference	0.016	0.008	0.039
	Home factor	0.348	0.459	0.448
	Ground effect	−3.265	1.519	0.032
Loss/draw (−1/0)	Intercept	1.967	0.680	0.004
	Rating difference	−0.021	0.008	0.007
	Home factor	−1.044	0.511	0.041
	Ground effect	−3.761	1.603	0.019

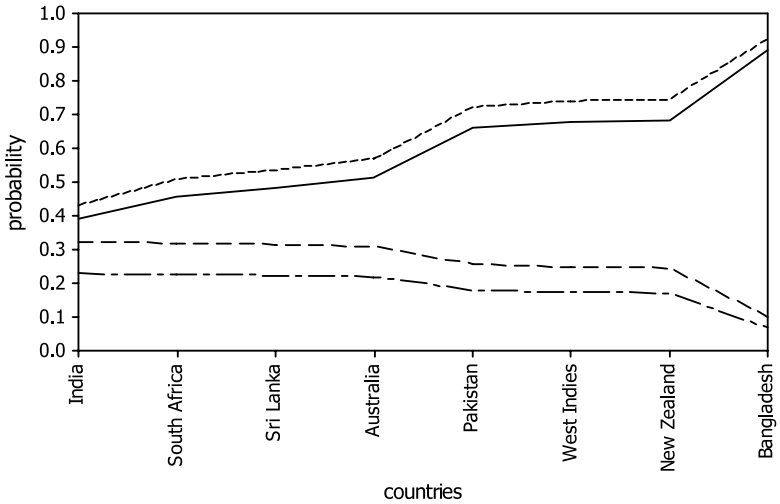

Fig. 2. Win and draw probabilities for the English cricket team in hypothetical matches against various teams at two grounds (Lord's and Headingley), using the best fitting model at the start of the match, and based on October 2010 ICC ratings. ------ probability of win at Headingley; ___ probability of win at Lord's; _ _ _ _ probability of draw at Lord's; _._._ probability of draw at Headingley.

England win probability is considerably lower at Lord's than at Headingley. To investigate the effect of the batting order, we calculate from our fitted model the England win probabilities in both situations (i.e., batting first and batting second) at Headingley when playing hypothetical matches against the rest of the current ICC member teams. These probabilities (see Fig. 3) suggest that batting first only increases the probability of winning when playing against high rated teams.

4.2. Models for subsequent sessions

Multinomial logistic regression models are considered at the start of each session. The best fitting models are given in Table 5. In the day-one models, the lead, wicket resources used by the reference team, home factor, ground effect and team strength were found to be important (based on the AIC and Nagelkerke's R^2). In the day-two and day-three models, wicket resources used by the opponents was also found to be an important covariate. The explanatory power of the rating difference, ground effect and home factor decreases with time. The home factor and ground effect become insignificant by tea on the fourth day, and the effect of the rating difference is likewise insignificant by the start of the fifth day. The effects of these factors broadly translate into lead and wicket resources used, so that lead and wicket resources used dominate by the start of the fifth day, as we would expect. The

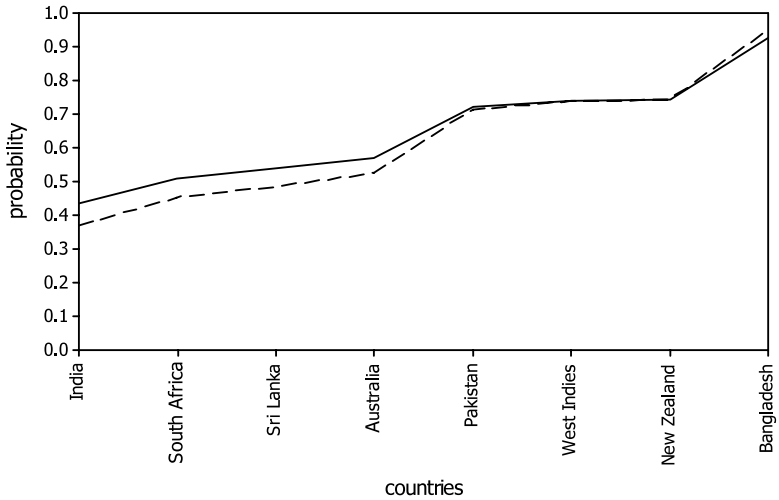

Fig. 3. Win probabilities for the English cricket team in hypothetical matches against various teams at Headingley, using the best fitting model at the start of the match, and based on October 2010 ICC ratings; ___ probability of a win at Headingly when England batted first; ------ probability of a win at Headingley when England batted second.

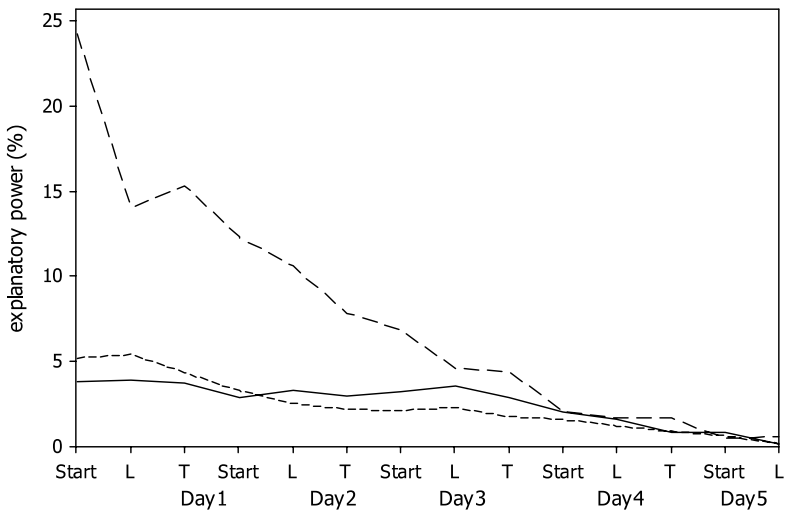

Fig. 4. Session by session, the explanatory power of the rating difference _ _ _, home factor ------, and ground effect ___.

change over time in the explanatory power of the pre-match factors is illustrated in Fig. 4. The explanatory power of an individual covariate is calculated as the difference between Nagelkerke's R^2 for the best fitting model, for the session, with and without the particular covariate.

In addition, we used the run-rate as a covariate for capturing the present quality of the batting conditions, but its effect was found to be insignificant. Note that we have ignored quadratic and higher order terms in the model fitting due to the limited number of observations in our dataset. Other factors which could also influence the match result, such as the bowling and batting strengths, and weather conditions, are not quantified in our data. Captains would be expected to take these factors into account when considering their batting and bowling strategies in each session.

The detailed results for a selection of the models are presented in the following sections.

4.3. Start of second day

The nominal multinomial logistic regression models considered at the start of the day-two position are given in Table 6. At the start of the second day, the lead, team strength, home factor, and wicket resources used by the reference team were found to be important (based on the AIC and Nagelkerke's R^2). The concordance table (Table 7) shows that the model predicts 68.5% of match outcomes correctly. This shows that the model provides good predictions at this stage of the match, to a certain extent. These prediction accuracy measures are based on in-sample forecasts, and should thus be treated with caution. We felt that there were not sufficient matches in the data set to enable out-of-sample forecasts. On the other hand, we might argue that we are not over-fitting the models, given that the covariate sets for the best models are as we might have chosen *a priori*. Estimates for the

Table 5
Best fitting multinomial logistic regression model at each stage, and showing Nagelkerke's R^2, the percentage of correct match outcome predictions (within the sample), the explanatory powers of the ground effect, the rating difference and the home factor, and the number of matches in the sample. The covariates here are L, the lead of reference team; RD, the ICC rating difference; H, the home factor; G, the ground effect; WR_1, the wicket resources used by the reference team; and WR_2, the wicket resources used by the opponents.

Day	Session	Model	Nag.R^2 (%)	Correct predictions (%)	Exp.power G (%)	Exp.power RD (%)	Exp.power H (%)	Number of matches
Day 1	Start of match	RD + H + G	37.0	59.6	3.9	24.3	5.2	146
	At lunch	RD + H + L + G + WR_1	43.8	63.7	3.9	14.1	5.4	146
	At tea	RD + H + L + G + WR_1	50.0	66.4	3.8	15.3	4.4	146
Day 2	Start of day	RD + H + L + G + WR_1	55.8	68.5	2.9	12.3	3.3	146
	At lunch	RD+H+L+G+WR_1+WR_2	57.4	71.2	3.3	10.6	2.5	146
	At tea	RD+H+L+G+WR_1+WR_2	62.6	75.3	3.0	7.9	2.2	146
Day 3	Start of day	RD+H+L+G+WR_1+WR_2	65.0	78.1	3.3	6.9	2.1	146
	At lunch	RD+H+L+G+WR_1+WR_2	67.8	75.3	3.6	4.6	2.3	146
	At tea	RD+H+L+G+WR_1+WR_2	73.2	80.1	2.9	4.4	1.8	146
Day 4	Start of day	RD+H+L+G+WR_1+WR_2	77.3	78.3	2.1	2.2	1.7	143
	At lunch	RD+H+L+G+WR_1+WR_2	74.9	80.2	1.6	1.7	1.2	131
	At tea	RD + L + WR_1 + WR_2	76.8	79.5	0.9	1.7	0.9	122
Day 5	Start of day	L + WR_1 + WR_2	79.9	81.1	0.8	0.5	0.6	111
	At lunch	L + WR_1 + WR_2	83.7	86.8	0.2	0.6	0.2	91
	At tea	L + WR_1 + WR_2	95.2	96.1	–	–	–	76

Table 6
Start of day-two: results of model fitting the multinomial logistic regression model to 146 test match outcomes for various sets of predictors: log-likelihood, the AIC and Nagelkerke's R^2. The covariates here are L, lead; RR, average run-rate of first two sessions; WR_1, wicket resources used of reference team; H, home factor; and G, ground effect.

Model	Parameters	LL	AIC	Nag.R^2 (%)
RD + H + L	8	−117.54	251.08	46.10
RD + H + G + L	10	−114.62	249.23	48.75
RD + H + L + WR_1	10	−109.74	239.48	52.93
RD + H + G + L + WR_1	**12**	**−106.18**	**236.35**	**55.81**
RD + H + G + L + WR_1 + WR_2	14	−104.56	237.13	57.07
RD + H + G + L + WR_1 + RR	14	−105.86	239.72	56.06
RD + H + G + L + WR_1 (ordinal)	7	−121.95	257.896	41.90

Table 7
Start of day-two: concordance table (cross-classification of the observed and predicted match outcomes) for the model shown in bold in Table 6. The predicted outcome is that with the highest probability.

Observed	Predicted			
	1	0	−1	Percent correct
1	49	6	8	77.80%
0	16	12	5	36.40%
−1	9	2	39	78.00%
Overall percentage	50.70%	13.70%	35.60%	68.50%

emboldened (best) model are in Table 8. The win and loss probabilities for certain specified covariate values (for lead, rating difference, home factor, and wicket resources used by the reference team) are shown in a later example in Section 5 (Figs. 7 and 8) for various values of lead and wicket resources.

4.4. At lunch on third and fourth days

At lunch on the third and fourth days, the lead, rating difference, home factor, ground effect and wicket resources used of both teams were found to be important (based on the AIC and Nagelkerke's R^2). Concordance tables are used for evaluating the predictive accuracy of the best fitted model (Tables 9 and 11). Table 9 shows that the model predicts 75.3% of the cases correctly, while Table 11 shows 80.2%. The maximum likelihood estimates for the best models are given in Tables 10 and 12. Table 10 indicates that the win-draw probability ratio depends strongly on the opponents' wicket resources used, whereas the loss-draw probability ratio depends strongly on both the current lead at the start of session and the ground effect. On the other hand, Table 12 shows that the loss-draw probability ratio depends strongly on both the current lead and the wicket resources used by the reference team. The win-draw probability ratio depends strongly on the wicket resources used by the opponents only.

5. Forecasting match outcome

The objective of this paper is to provide a quantitative means of forecasting match outcomes in play. The quality of these forecasts is summarised in Table 5. As the match

Table 8
Start of day-two: fitted parameter estimates for the minimum AIC logistic regression model with the covariates: lead, rating difference, wicket resources used by the reference team, and home factor, with standard errors and *p*-values, 146 test matches.

		Coefficient	s.e	*p*-value
Win/draw (1/0)	Intercept	2.071	1.319	0.116
	Rating difference	0.018	0.008	0.020
	Home factor	0.437	0.477	0.360
	Ground effect	−3.300	1.530	0.031
	Lead	−0.006	0.004	0.100
	Wicket resources used (WR_1)	0.015	0.009	0.122
Loss/draw (−1/0)	Intercept	1.873	1.557	0.229
	Rating difference	−0.018	0.009	0.042
	Home factor	−1.062	0.630	0.092
	Ground effect	−4.333	1.892	0.022
	Lead	−0.013	0.004	0.001
	Wicket resources used (WR_1)	0.048	0.013	0.000

Table 9
At lunch on day-three: concordance table (cross-classification of the observed and predicted match outcomes) for the best fitting model. The predicted outcome is that with the greatest probability.

Observed	Predicted			
	1	0	−1	Percent correct
1	50	6	7	79.40%
0	10	18	5	54.50%
−1	5	3	42	84.00%
Overall percentage	44.50%	18.50%	37.00%	75.30%

Table 10
At lunch on day-three: fitted parameter estimates for the minimum AIC logistic regression model (nominal) with covariates: the lead, rating differences; WR_1, wicket resources of the reference team used; WR_2, wicket resources of opponents used; home factor; and ground effect; with standard errors and *p*-values, 146 test matches (Nov. 2005–March 2010).

		Coefficient	s.e	*p*-value
Win/draw (1/0)	Intercept	−2.560	1.941	0.187
	Rating difference	0.009	0.009	0.274
	Home factor	0.667	0.517	0.197
	Ground effect	−4.279	1.674	0.011
	Lead	0.002	0.002	0.277
	Wicket resources used (WR_1)	0.014	0.019	0.449
	Wicket resources used (WR_2)	0.032	0.011	0.004
Loss/draw (−1/0)	Intercept	−0.637	2.202	0.772
	Rating difference	−0.018	0.010	0.067
	Home factor	−0.935	0.708	0.187
	Ground effect	−5.918	2.178	0.007
	Lead	−0.010	0.002	0.000
	Wicket resources used (WR_1)	0.036	0.021	0.090
	Wicket resources used (WR_2)	0.008	0.014	0.587

progresses, the ability of the models to forecast correctly increases (from 59.6% at the start of the match to 96.1% at tea on the fifth day).

It is interesting to compare our model forecasts with forecasts based on bookmaker odds. Pre-match and in-play betting odds on test match outcomes are offered in real time by bookmakers and betting sites. Oddschecker (http://www.oddschecker.com/cricket/) provides a convenient summary of these, although, as the odds are only available in play, they are laborious to collect. Consequently, we collected session by session odds for only a handful of matches, two of which are presented here as examples. The odds offered by 15 bookmakers were obtained; these odds were converted into probabilities by accounting for the over-round, and then averaged over the bookmakers. The session by session scorecard for the second test between Sri Lanka and India in 2010 is presented in Table 13. The model forecasts, based on the best fitting models in Table 5, are then compared with the bookmaker forecasts in Fig. 5. We can see that the model and bookmaker forecasts are broadly similar, but that the model forecast rather overestimates the Sri Lanka win probability and underestimates the draw probability, particularly at tea on day three when India's first innings was in progress and they were trailing by 399 runs. If we accept that the bookmaker odds represent the match situation well, then perhaps the model here was less able to capture the effects of the pitch condition or the Sri Lanka bowling or India batting strengths. In a second example, the first test between England and Pakistan, 2010 (Fig. 6), the model appears to do better, particularly later in the match. It should be noted that some bookmakers take the batting order (toss advan-

Table 11
At lunch on day-four: concordance table (cross-classification of the observed and predicted match outcomes) for the best fitting model. The predicted outcome is that with the greatest probability.

Observed	Predicted			
	1	0	−1	Percent correct
1	53	3	3	89.80%
0	9	20	4	60.60%
−1	4	3	32	82.10%
Overall percentage	50.40%	19.80%	29.80%	80.20%

Table 12
At lunch on day-four: fitted parameter estimates for the minimum AIC logistic regression model (nominal) with the covariates: lead, rating differences, WR_1 (wicket resources of reference team used), WR_2 (wicket resources of opponents used), ground effect and home factor with standard errors and *p*-values, 146 test matches (Nov. 2005–March 2010).

		Coefficient	s.e	*p*-value
	Intercept	−9.293	2.551	0.000
	Rating difference	0.005	0.009	0.563
	Home factor	0.413	0.609	0.497
Win/draw (1/0)	Ground effect	−2.951	1.727	0.088
	Lead	0.003	0.002	0.195
	Wicket resources used (WR_1)	0.011	0.001	0.256
	Wicket resources used (WR_2)	0.081	0.025	0.001
	Intercept	−7.002	2.479	0.005
	Rating difference	−0.011	0.011	0.277
	Home factor	−1.019	0.781	0.192
Loss/draw (−1/0)	Ground effect	−4.293	2.161	0.047
	Lead	−0.012	0.003	0.000
	Wicket resources used (WR_1)	0.051	0.014	0.000
	Wicket resources used (WR_2)	0.032	0.024	0.189

Table 13
Example 1: scorecard for the 2nd test between Sri Lanka (reference team) and India played at Sinhalese Sports Club Ground, Colombo, Sri Lanka in 2010. W_1: reference team wickets down. W_2: other team total wickets down.

Session	Day 1		Day 2			Day 3			Day 4			Day 5		
	At lunch	At tea	Start	At lunch	At tea	Start	At lunch	At tea	Start	At lunch	At tea	Start	At lunch	At tea
Lead	128	235	312	457	587	547	469	399	260	165	53	−27	−46	33
W_1	1	1	2	2	3	10	10	10	10	10	10	10	10	13
W_2	0	0	0	0	0	0	3	4	4	4	5	9	10	10

tage) into account and some do not; and therefore the odds at the start of the match do not provide a fair comparison with the model forecasts.

The model forecasts might also be used in the following way. Consider the 2010–2011 "Ashes", a five-match series played between England and Australia, with a history of more than 100 years. At the time of writing, the series is about to start, with Australia holding the home field advantage, but with England having a slightly higher ICC rating. This will be an interesting series for all stakeholders: fans, teams, and communication-media businesses. The first match of the series is to take place in Brisbane, and, using model estimates in Table 8, we forecast match outcomes at the start of the second day for various hypothetical positions. From these forecasts we can deduce what would be good and poor outcomes for the teams on the first day, for example. Thus, if England bat first (Fig. 7), the match will be evenly poised if England are 290/6 or 350/8 at the start of day 2. On the other hand, if Australia bat first (Fig. 8), the match will be evenly poised if Australia score 190/8. Thus the model indicates that England would need to bowl Australia out on the first day to put themselves into a leading position. Other broad effects are evident in Figs. 7 and 8. The probability of a win increases with the lead and wickets remaining, up to a point; if the lead is large and the number of wickets remaining is large, then the win probability is lower, as a draw is more likely. This is the case for both teams, although the win probability for Australia is higher overall given their home advantage, particularly if they bat first.

In summary then, although we were not able to systematically compare bookmaker odds with model forecasts, the two examples presented indicate that the model forecasts are not unreasonable. Whether the models could provide an opportunity to exploit inefficiencies in the betting market is a question which requires further study. However, where the models may be useful is in providing team management with a tool for monitoring the strength of their position and for exploring match position scenarios quickly and objectively.

6. Discussion

The purpose of this paper is to model the match outcome at the start of each session in test cricket. Looking at the start of session positions has two benefits. Firstly, some progress can be made with the quantitative analysis of the problem. Secondly, the models can guide the team

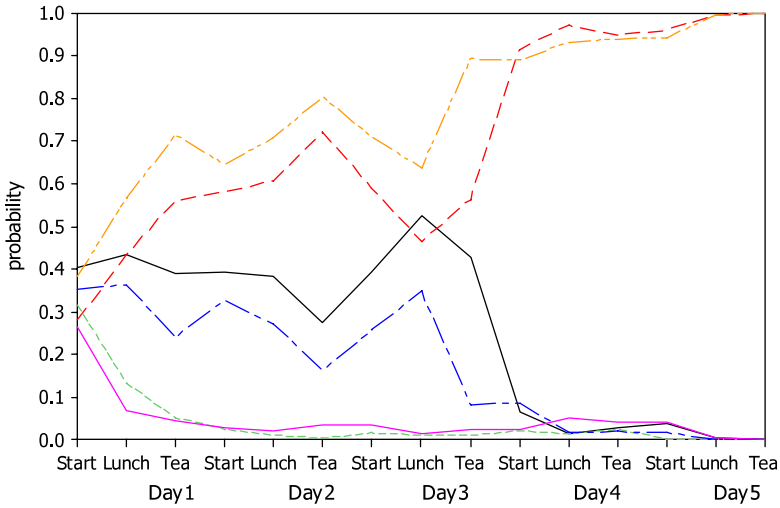

Fig. 5. Probability forecasts based on our models: ___ Sri Lanka win (black); _ _ _ _ draw (red); ------ India win (green). Probability forecasts based on bookmaker odds: _._._ Sri Lanka win (blue); _.._.._ draw (orange); ___ India win (pink). Sri Lanka (reference team) vs. India, 2010, Sinhalese Sports Club Ground, Colombo, Sri Lanka. Match drawn (Online figure is in color).

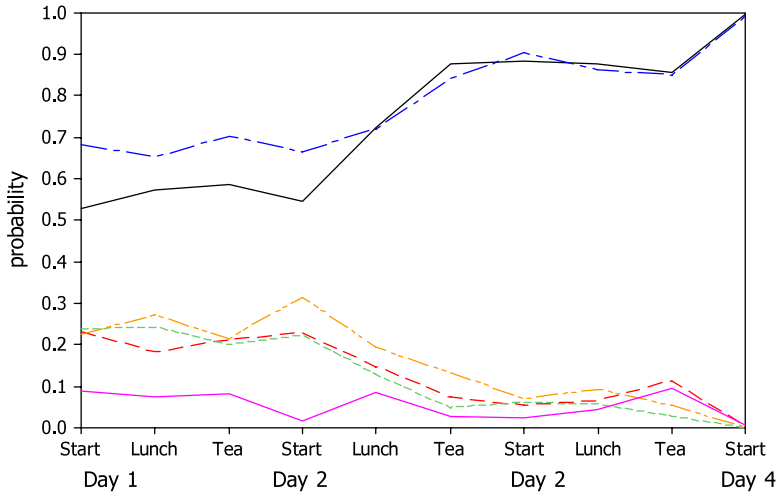

Fig. 6. Probability forecasts based on our models: _._._ England win (blue); _.._.._ draw (orange); ___ Pakistan win (pink). Probability forecasts based on bookmaker odds: ___ England win (black); _ _ _ _ draw (red); ------ Pakistan win (green). England (reference team) vs. Pakistan, 2010, Trent Bridge, Nottingham. England won by 354 runs (Online figure is in color).

captains and management with respect to batting and bowling strategies in each session. With $X(t)$ denoting the match state at time t, Y the match outcome, and S the strategy adopted in (t_0, t_1), the strategic decision problem can be stated as follows: first determine $\text{prob}(Y|X(t))$ (this is the model which we develop in this paper); then determine $\text{prob}(X(t_1)|X(t_0), S)$ $(t_0 < t_1)$; then compute $\text{prob}(Y|X(t_0), S) = \text{prob}(Y|X(t_1)) \times \text{prob}(X(t_1)|X(t_0), S)$ for various $S = s$ in (t_0, t_1); finally, choose the value of S which "optimizes" $\text{prob}(Y|X(t_0), S)$. This problem is straightforward to state. However, the implementation is difficult, because S is generally unobserved, and thus $\text{prob}(X(t_1)|X(t_0), S)$ is difficult to quantify. One solution is to explore different $X(t_1)$ scenarios (which are plausible given $X(t_0)$ and S) by considering $\text{prob}(Y|X(t_1))$ and the subjective probability of the decision maker about the transition from $X(t_0)$ to $X(t_1)$ if he adopts strategy $S = s$ in the period (t_0, t_1). The model we develop in this paper facilitates this approach, and the third example above relating to the Ashes test in Brisbane in 2010 illustrates in a sense how this analysis might be presented. From a less technical viewpoint, the graphical presentation of probabilities $\text{prob}(Y|X(t))$ might also be of interest to the media and viewing public. A generalization of the methodology to other sports is possible in principle, although the utility of the approach is questionable where a sport has few natural breaks (e.g. soccer, hockey, American football). However, an over-by-over analysis of the twenty-overs form of cricket might be amenable.

Regarding particular characteristics of the model $\text{prob}(Y|X(t))$, our results show the way in which the covariates which influence the match outcome vary from

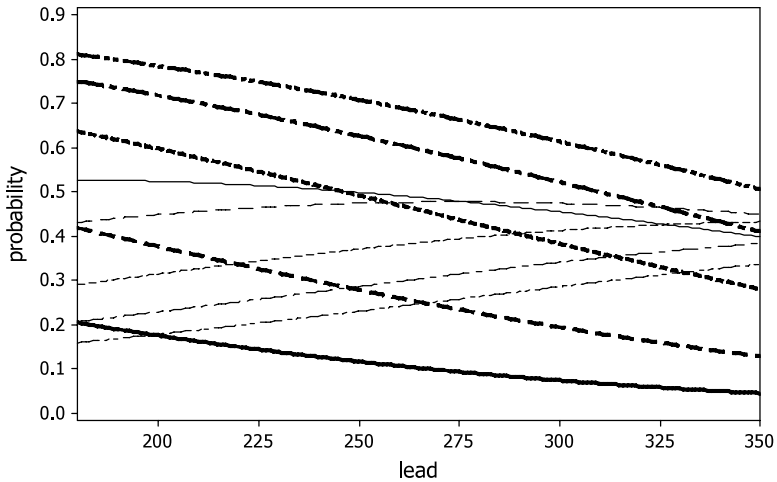

Fig. 7. England win (normal lines) and loss (bold lines) probabilities at the position at the start of the second day with England batting first, as a function of the lead established and wickets down: ___ 2 wickets down; ----- 4 wickets down ..... 6 wickets down; __.__. 8 wickets down; _.._.. 10 wickets down; home factor, H = 0; rating difference, RD = 2; ground effect (Brisbane), G = 0.21.

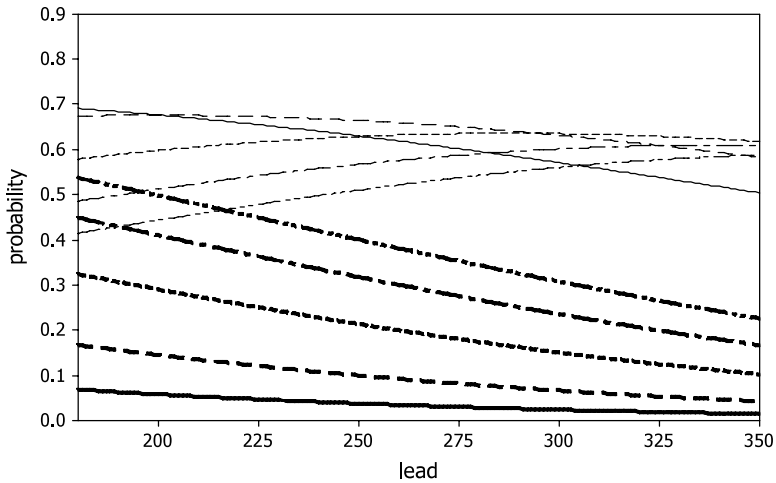

Fig. 8. Australia win (normal lines) and loss (bold lines) probabilities at the position at the start of the second day with Australia batting first, as a function of the lead established and wickets down: ___ 2 wickets down; ----- 4 wickets down; ..... 6 wickets down; __.__. 8 wickets down; _.._.. 10 wickets down; home factor, H = 1; rating difference, RD = −2; ground effect (Brisbane), G = 0.21.

session to session. Early in the match, pre-match team strengths have a large effect. The home advantage is small, while the ground effect (tendency for a draw at a particular ground) is large. These effects decrease as the match progresses. This is because they translate into lead and wicket resources as a match progress, so that later in a match, lead and wicket resources dominate. Of course, other factors are also influential, such as the weather conditions, and bowling and batting strengths. One of the limitations of our work is that if both teams lose wickets in the first day's play then our models are not able to take the number of wickets down into account for the team batting second. Data on more matches would helpful for overcoming this problem.

As we would expect, the forecasting accuracy of the model improves as the match progresses. It would be interesting to carry out a study comparing the accuracy of the forecasts from the model based on the match state with those based on betting odds. However, we were not able to do this as we did not have such odds histories for a sufficient number of matches. It would also be interesting to consider models which were fitted sequentially rather than independently, so that the linear predictor from the model at one stage is a covariate in the model at the following stage. However, this would be another study. Finally, a potential application of the in-play forecasts which we calculate would be a rating of the batting, bowling and fielding contributions of players, based on the effects of a player's contributions on the predicted match outcome, in the manner proposed by Johnston, Clarke, and Noble (1993) for limited-overs cricket; Scarf, Akhtar, and Rasool (2011) explore such a rating system in test cricket.

This type of analysis of player ratings would be in the spirit of that developed for soccer by McHale, Scarf, and Folker (in press) and Scarf and McHale (2005). Lewis (2005, 2008) also suggested a similar rating system for limited-overs cricket (50 overs), based upon resources used.

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